Bending Without Tearing

A topological account of two deformation programs:
circle $\to$ square and donut $\to$ mug

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Abstract

Two small C++ programs deform one shape into another: the first turns a circle into a square, the second a donut into a coffee mug. This note explains both in the language of topology. The first program implements an explicit homeomorphism of the plane, the radial change of norm $F(p) = (\|p\|_2/\|p\|_\infty)\, p$, together with a straight-line ambient isotopy; we prove that every stage is a homeomorphism and that the corners of the square are invisible to topology. The second program has no closed formula. It builds a one-parameter family of solids S_t from an exact solid torus ($t = 0$) to a mug ($t = 1$) and monitors their Betti numbers, computed from the Euler characteristic and the connected components of a cubical complex on a voxel grid. We show that a donut and a mug are both a ball with one handle, that a dent is not a hole, which events would change the Betti numbers, and that the program's algorithm computes them exactly for its voxel model. Control experiments whose outcomes are predicted by the Mayer–Vietoris sequence agree with the computation. We state precisely what is proved, what is classical, and what is verified only numerically.

1 Introduction

Topologists are said to be unable to tell a doughnut from a coffee cup. This note takes the joke seriously. It explains, in the language of topology, what two small C++ programs do when they "turn a circle into a square" and "turn a donut into a mug", and it is explicit about what those programs establish and what they only suggest.

The two programs differ in kind.

Program I (`circle_to_square.cpp`) implements an *explicit homeomorphism* of the plane together with a straight-line *ambient isotopy*. Everything it does can be proved in a few lines of analysis (Section 3); the checks the program prints are sanity checks of the floating-point implementation of a statement that is already a theorem.

Program II (`donut_to_mug.cpp`) has no closed formula to lean on. It builds a one-parameter family of solids S_t, $t \in [0, 1]$, that starts at an exact solid torus and ends at a mug, and it *measures* topological invariants—the Betti numbers—of every stage on a voxel model (Section 4). Its logic is a mixture: structural reasons (both ends are "a ball with one handle") explain why the answer should not change, a lemma says exactly which events would change it, and computation tests whether such an event occurs along the family.

Contributions.

(1) We prove that the radial change of norm F is a homeomorphism of the plane carrying every circle onto a square, that the straight-line deformation H_t is an ambient isotopy, and that the corners of the square are invisible to topology (Section 3).

(2) We explain why a donut is a mug in the language of handle decompositions, with two lemmas: a dent is a graph deformation and cannot create a hole; and two contractible pieces glued along two contractible pieces have the homology of a solid torus (Sections 4.1 to 4.4).

(3) We describe precisely the family S_t used by the program and catalogue the events that would change its topology (Sections 4.5 and 4.6).

(4) We prove that the program's voxel algorithm computes the exact Betti numbers of the cubical complex it builds (Section 4.7).

(5) We report the computed Betti numbers along the family, together with control experiments whose outcomes are predicted by the lemmas, and we state what the evidence does not show (Sections 4.8 and 4.9).

Status labels. Statements called Proposition, Lemma or Theorem are proved here or are classical (with a reference). Statements called Observation are numerical facts obtained by running the programs, or small harnesses around them; they are evidence, not proofs. Every number quoted in this note was produced by running the code; none is typed from memory. Section 6 collects the status of every claim in one ledger.

2 Preliminaries

2.1 Spaces, maps and invariants

All spaces are subspaces of $\mathbb{R}^n$ with the subspace topology. A map $f\colon X \to Y$ is a *homeomorphism* if it is a continuous bijection whose inverse is continuous; we then write $X \cong Y$. A *topological invariant* is a property or number that is the same for homeomorphic spaces. We use compactness; the number of connected components; the *Betti numbers* $\beta_k(X) = \operatorname{rank} H_k(X;\mathbb{Z})$; and the *Euler characteristic* $\chi(X) = \sum_k (-1)^k \beta_k(X)$. Lengths, angles, curvature and "having corners" are not topological invariants.

For a compact $X \subset \mathbb{R}^3$, β_0 is the number of pieces, β_1 is the number of independent tunnels (loops in X that cannot be shrunk inside X) and β_2 is the number of sealed hollow cavities; Theorem 4.9 makes the last statement precise. Table 1 lists the values used below.

2.2 Deforming through time

Definition 2.1. An *isotopy* of a space $X \subset \mathbb{R}^n$ is a continuous map $H\colon X \times [0,1] \to \mathbb{R}^n$ such that every $H_t = H(\cdot, t)$ is a homeomorphism onto its image. An *ambient isotopy* is a continuous map $\Phi\colon \mathbb{R}^n \times [0,1] \to \mathbb{R}^n$ with $\Phi_0 = \mathrm{id}$ such that every Φ_t is a homeomorphism of $\mathbb{R}^n$; it carries a set A to the sets $A_t = \Phi_t(A)$.

Table 1. Betti numbers and Euler characteristic of some basic spaces.

space	β_0	β_1	β_2	χ
circle S^1, or a square outline	1	1	0	0
closed disc, or a filled square	1	0	0	1
solid ball D^3	1	0	0	1
solid torus $D^2 \times S^1$ (donut, mug)	1	1	0	0
solid ball with a sealed cavity	1	0	1	2
two disjoint solid balls	2	0	0	2
solid double torus (two handles)	1	2	0	-1

Remark 2.2. If $A_t = \Phi_t(A)$ then every $A_t \cong A$. The converse is false: a knotted and an unknotted circle in $\mathbb{R}^3$ are homeomorphic but not ambient isotopic. Thus "the donut is the mug" (a statement about two spaces) is weaker than "the donut can be deformed into the mug inside $\mathbb{R}^3$" (a statement about two embeddings), and an animation shows the second.

2.3 Handles

A *0-handle* is a closed 3-ball. Let N be a compact 3-manifold and $D_1, D_2 \subset \partial N$ disjoint closed discs. *Attaching a 1-handle* to N means gluing a copy of $D^2 \times [0,1]$ to N by identifying $D^2 \times \{0\}$ with D_1 and $D^2 \times \{1\}$ with D_2. Attaching a 1-handle to a ball produces, when the result is orientable, the solid torus $D^2 \times S^1$ (Proposition 4.3). In words: *a solid torus is a ball with one handle*. See Hempel [2] for background on 3-manifolds.

3 Program I: the circle and the square

3.1 The objects and the program

Write $p = (x, y) \in \mathbb{R}^2$, $\|p\|_2 = \sqrt{x^2 + y^2}$ and $\|p\|_\infty = \max(|x|, |y|)$, so that $\|p\|_\infty \le \|p\|_2 \le \sqrt{2}\,\|p\|_\infty$. Let

$$S^1 = \{\|p\|_2 = 1\}, \qquad \Sigma = \{\|p\|_\infty = 1\},$$
$$D = \{\|p\|_2 \le 1\}, \qquad Q = \{\|p\|_\infty \le 1\} = [-1,1]^2$$

be the circle, the outline of the square, the closed disc and the filled square. For $p \neq 0$ put $\lambda(p) = \|p\|_2 / \|p\|_\infty \in [1, \sqrt{2}]$ and define

$$F(p) = \lambda(p)\, p, \qquad G(q) = \frac{q}{\lambda(q)}, \qquad F(0) = G(0) = 0. \tag{1}$$

On S^1 one has $F(p) = p/\|p\|_\infty$, which is the formula in the program (function `circleToSquare`); on Σ one has $G(q) = q/\|q\|_2$ (`squareToCircle`). The animation is the straight-line deformation (`morph`)

$$H_t(p) = (1-t)\, p + t\, F(p) = \lambda_t(p)\, p, \qquad \lambda_t = 1 - t + t\lambda, \qquad t \in [0,1]. \tag{2}$$

```
Pt circleToSquare(Pt p) {                          // F, abridged
    double m = max(fabs(p.x), fabs(p.y));          // the max-norm of p
    if (m == 0.0) return {0.0, 0.0};
    double k = hypot(p.x, p.y) / m;                // lambda(p)
    return {p.x * k, p.y * k};
}
Pt morph(Pt p, double t) {                         // H_t
    Pt q = circleToSquare(p);
    return {(1 - t) * p.x + t * q.x, (1 - t) * p.y + t * q.y};
}
```

3.2 F is a homeomorphism

Proposition 3.1. *The maps F and G of (1) are mutually inverse homeomorphisms of $\mathbb{R}^2$. For every $r \geq 0$, F maps the Euclidean circle $\{\|p\|_2 = r\}$ onto the square outline $\{\|q\|_\infty = r\}$; in particular $F(S^1) = \Sigma$ and $F(D) = Q$.*

Proof. The function λ is continuous on $\mathbb{R}^2 \setminus \{0\}$ (a quotient of continuous functions with non-vanishing denominator) and constant along rays: $\lambda(cp) = \lambda(p)$ for $c > 0$. Hence F and G are continuous on $\mathbb{R}^2 \setminus \{0\}$. At the origin, $\|F(p)\|_2 = \lambda(p)\|p\|_2 \leq \sqrt{2}\|p\|_2$ and $\|G(q)\|_2 \leq \|q\|_2$, so both are continuous there too. Since $F(p)$ is a positive multiple of p, $\lambda(F(p)) = \lambda(p)$, and therefore $G(F(p)) = F(p)/\lambda(p) = p$; the same argument gives $F(G(q)) = q$. So F is a bijection with continuous inverse G. Finally $\|F(p)\|_\infty = \lambda(p)\|p\|_\infty = \|p\|_2$, so F maps $\{\|p\|_2 = r\}$ into $\{\|q\|_\infty = r\}$, and G maps it back; hence F maps onto. □

3.3 The straight-line deformation is an ambient isotopy

Proposition 3.2. *The map $H\colon \mathbb{R}^2 \times [0,1] \to \mathbb{R}^2$ of (2) is continuous, $H_0 = \mathrm{id}$, $H_1 = F$, and every H_t is a homeomorphism of $\mathbb{R}^2$ with inverse $K_t(q) = q/\lambda_t(q)$. Thus H is an ambient isotopy from the circle S^1 to the square outline Σ, and from the disc D to the filled square Q.*

Proof. Continuity of H is clear away from the origin, and $\|H_t(p)\|_2 \leq \sqrt{2}\|p\|_2$ gives continuity at the origin uniformly in t. Because $1 \leq \lambda_t \leq \sqrt{2}$ and λ_t is constant along rays, the argument of Proposition 3.1 applies verbatim: $\lambda_t(H_t(p)) = \lambda_t(p)$, so $K_t(H_t(p)) = p$ and $H_t(K_t(q)) = q$, and K_t is continuous. □

Corollary 3.3. *For every $t \in [0,1]$ the curve $A_t = H_t(S^1)$ is a simple closed curve. In polar coordinates it is the graph $\rho = 1 - t + t/\max(|\cos\theta|, |\sin\theta|)$; in particular it is star-shaped about the origin. A_0 is the circle and A_1 the square outline.*

3.4 What the program checks

The program samples 10^5 points p on S^1 and q on Σ and reports the largest values of $\big|\|F(p)\|_\infty - 1\big|$, of $\|G(F(p)) - p\|$ and of $\|F(G(q)) - q\|$. The printed values are 2.2×10^{-16}, 2.5×10^{-16} and 4.0×10^{-16}, that is, rounding error. It also forms, for $t = 0, 0.1, \ldots, 1$, the 360-gon with vertices $H_t(e^{2\pi ik/360})$ and verifies that consecutive vertices turn counter-clockwise through positive angles adding up to 2π (function `windsOnce`). A closed polygon whose vertices appear in

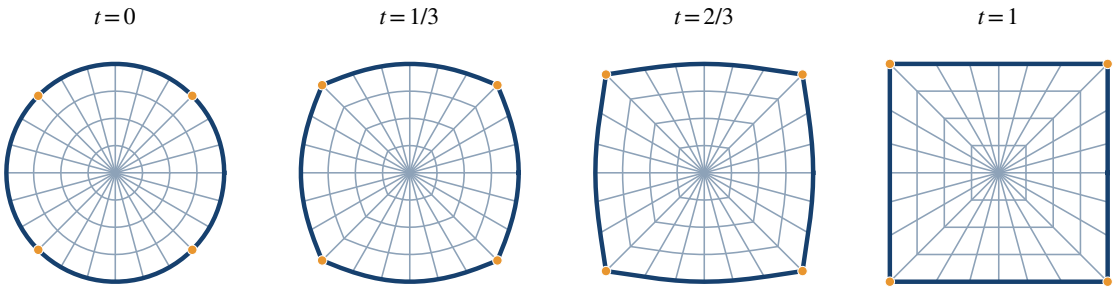


Figure 1. The deformation H_t applied to a polar grid (rings of radius $1/4, 1/2, 3/4, 1$ and 24 spokes). The bold curve is $H_t(S^1)$. The four orange points are the images of the points of S^1 on the diagonals $|x| = |y|$; they become the corners of the square and are the only places where F fails to be smooth. Because every H_t is a homeomorphism of the whole plane, the grid is bent but never torn, folded or glued.

strictly increasing angular order around an interior point is star-shaped about that point and hence simple; this is the discrete shadow of Corollary 3.3. All tests passed.

These checks test the arithmetic, not the mathematics. Continuity cannot be certified by sampling; it is established by Propositions 3.1 and 3.2.

3.5 Corners, smoothness and a generalization

Remark 3.4 (Corners). F is smooth on $\mathbb{R}^2$ minus the diagonals $\{|x| = |y|\}$, where $\|\cdot\|_\infty$ is, up to sign, one of the coordinates, but its differential jumps across the diagonals; the four points of S^1 on the diagonals are sent to the corners $(\pm 1, \pm 1)$. No diffeomorphism of the plane carries S^1 onto Σ, because Σ is not a C^1 curve. Topology is blind to this: it records which points are near which, not angles. A circle has no corners and a square has four, yet as topological spaces—and, by Proposition 3.2, as subsets of the plane up to ambient isotopy—they are the same.

Remark 3.5 (Generalization). Let $\Omega \subset \mathbb{R}^2$ be a compact region, star-shaped about 0, whose boundary is a polar graph $\rho = \rho_\Omega(\theta)$ with $\rho_\Omega > 0$ continuous. The same proof shows that $p \mapsto \rho_\Omega(\theta)\, p$ (θ the polar angle of p) is a homeomorphism of the plane carrying D onto Ω. Triangles, stars and blobs are therefore "the same" as the disc and their boundaries as the circle; the square is the case $\rho_Q(\theta) = 1/\max(|\cos\theta|, |\sin\theta|)$, for which $\rho_Q(\theta)\, p = \lambda(p)\, p = F(p)$.

4 Program II: the donut and the mug

Here there is no formula for a homeomorphism to prove correct. The program does three things: it fixes two end shapes (Section 4.1), joins them by a one-parameter family of shapes (Section 4.5), and measures the topology of every stage (Sections 4.7 and 4.8). The first four subsections below supply the theory that says what the measurements should be.

4.1 The two end shapes

Use coordinates (x, y, z) with z pointing up.

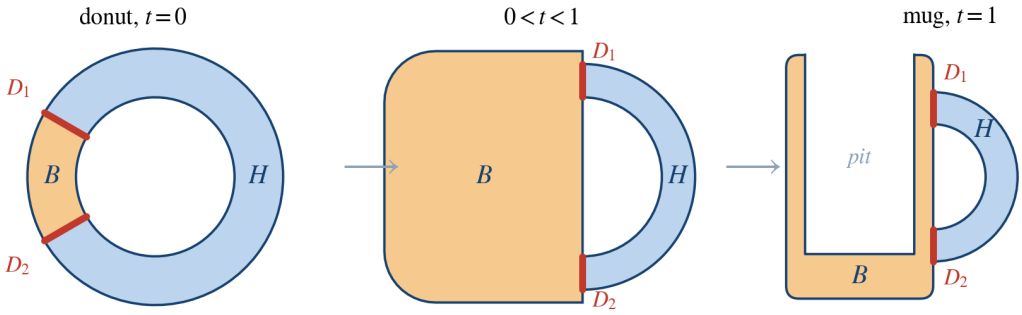


Figure 2. Schematic cross-sections of the common structure. Left: the donut is a ball B (a short piece of the ring, orange) with a handle H (the rest of the ring, blue) glued along two discs D_1, D_2 (red). Middle: during the deformation B swells while H becomes shorter and thinner. Right: the mug is again a ball B—the cup with its pit—with a handle H glued along two discs. The decomposition never changes; only the proportions do.

The donut. For $0 < r < R$ let $\Gamma = \{(R\cos\varphi, 0, R\sin\varphi)\}$ be a circle in the xz-plane, so that the hole of the donut points along y, and let

$$T = \{p \in \mathbb{R}^3 : \operatorname{dist}(p, \Gamma) \le r\}$$

(the program uses $R = 1$, $r = 0.36$). The map $E\big(ue^{i\psi}, \varphi\big) = \big((R + u\cos\psi)\cos\varphi,\ u\sin\psi,\ (R + u\cos\psi)\sin\varphi\big)$ is a continuous bijection from $D_r^2 \times S^1$ (D_r^2 the disc of radius r with polar coordinates (u, ψ)) onto T; the domain being compact and T Hausdorff, it is a homeomorphism. So $T \cong D^2 \times S^1$ is a solid torus.

The mug. Use cylindrical coordinates (ϱ, ϕ, z) about the vertical axis of the cup. Let $0 < R_{\text{in}} < R_c$ and $z_0 < z_f < z_1$ be the radii of the pit and of the cup and the heights of the base, the floor and the rim. The cup is

$$B = \{(\varrho, \phi, z) : 0 \le \varrho \le R_c,\ z_0 \le z \le u(\varrho)\}, \qquad u(\varrho) = \begin{cases} z_f & \varrho < R_{\text{in}}, \\ z_1 & R_{\text{in}} \le \varrho \le R_c, \end{cases}$$

a floor of thickness $z_f - z_0$, a wall of thickness $R_c - R_{\text{in}}$ and a pit of depth $z_1 - z_f$ cut into the top. The handle is the closed r_h-neighbourhood H of a circular arc whose endpoints lie inside the wall; let H' be the closure of $H \setminus B$, the part of the handle outside the cup. The mug is

$$M = B \cup H', \qquad B \cap H' = D_1 \sqcup D_2,$$

where D_1, D_2 are two disjoint discs on the outer wall (Figure 2, right). The rounded edges and fillets that the program adds are ignored in this section; Section 4.5 describes the version actually used.

4.2 A dent is not a hole

Lemma 4.1 (Dents are graph deformations). *Let $D \subset \mathbb{R}^2$ be a compact disc and $\ell < u$ continuous functions $D \to \mathbb{R}$. Then*

$$K(\ell, u) = \{(x, y, z) : (x, y) \in D,\ \ell(x, y) \le z \le u(x, y)\}$$

is homeomorphic to $D \times [0,1]$, a ball, by $\Psi(x,y,z) = \left(x, y, \frac{z-\ell}{u-\ell}\right)$. In particular, if $u_\tau = (1-\tau)u_0 + \tau u_1$ with $u_0, u_1 > \ell$, the sets $K(\ell, u_\tau)$, $\tau \in [0,1]$, form a continuous family of balls: pushing a dent into the top face cannot change the topology as long as the floor keeps positive thickness.

Proof. Ψ is continuous with continuous inverse $(x,y,s) \mapsto (x,y,\ell + s(u-\ell))$, because $u - \ell > 0$ on the compact set D. For u_τ note that $u_\tau - \ell = (1-\tau)(u_0 - \ell) + \tau(u_1 - \ell) > 0$. □

Remark 4.2 (Vertical walls; what would go wrong). In the program the inner wall of the pit is exactly vertical, so u has a jump and Lemma 4.1 does not apply verbatim. The conclusion survives: in cylindrical coordinates the cup is the solid of revolution of an L-shaped planar disc that meets the axis in a boundary arc, and such a solid is a ball; alternatively, replace the vertical wall by a very steep one. The one thing that does break the conclusion is $u - \ell \to 0$: if the pit's floor reached the base, the cup would become an open ring—a solid torus—and the mug would acquire a second tunnel. The program stops the pit at a floor thickness of $0.18 > 0$.

4.3 The donut is the mug

Proposition 4.3 (classical). *$T \cong M$. Consequently $\partial T \cong \partial M$, a 2-torus.*

Proof. We have $T \cong D^2 \times S^1$ by Section 4.1. For M: (i) the cup B is the region between two graphs over a disc, so $B \cong D^2 \times [0,1]$ by Lemma 4.1 and Remark 4.2; (ii) H' is a tube around an arc, so $H' \cong D^2 \times [0,1]$, with end discs $D_1, D_2 \subset \partial B$; (iii) by the Schoenflies theorem in the sphere ∂B and the Alexander trick, there is a homeomorphism $B \to D^2 \times [0,1]$ carrying D_1 and D_2 to the two end discs (see Moise [9]). Hence M is $D^2 \times [0,2]$ with its two ends identified by some homeomorphism g of D^2. Since $M \subset \mathbb{R}^3$ is orientable, g preserves orientation (an orientation-reversing g would give the solid Klein bottle, which does not embed in $\mathbb{R}^3$), and every orientation-preserving homeomorphism of the disc is isotopic to the identity (Alexander trick). So $M \cong D^2 \times S^1$. A homeomorphism of 3-manifolds with boundary restricts to a homeomorphism of the boundaries. □

Thus "the donut is the mug" means: both are a 0-handle with one 1-handle (Figure 2). The hole of the donut is the gap inside the handle of the mug; the cup is a ball, and a dent in a ball is not a hole.

4.4 Counting tunnels: two balls and two discs

Lemma 4.4 (Two contractible pieces, two contractible contacts). *Let $A, B \subset \mathbb{R}^3$ be compact polyhedra, each contractible, such that $A \cap B$ has exactly two connected components, each contractible. Then $A \cup B$ is connected with $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}$ and $H_k = 0$ for $k \geq 2$. Thus $(\beta_0, \beta_1, \beta_2) = (1,1,0)$ and $\chi = 0$.*

Proof. The reduced Mayer–Vietoris sequence of the subcomplexes A, B of $A \cup B$ (Hatcher [1, §2.2]) reads

$$\cdots \to \tilde{H}_k(A \cap B) \to \tilde{H}_k(A) \oplus \tilde{H}_k(B) \to \tilde{H}_k(A \cup B) \to \tilde{H}_{k-1}(A \cap B) \to \cdots$$

Since A and B are contractible the middle groups vanish, so $\tilde{H}_k(A \cup B) \cong \tilde{H}_{k-1}(A \cap B)$ for all k. For an intersection consisting of two contractible components, $\tilde{H}_0(A \cap B) \cong \mathbb{Z}$ and $\tilde{H}_j(A \cap B) = 0$ for $j \geq 1$, and $\tilde{H}_{-1}(A \cap B) = 0$ because $A \cap B \neq \varnothing$. Hence $\tilde{H}_1(A \cup B) \cong \mathbb{Z}$ and all other reduced groups vanish. □

The same computation predicts what happens when the hypotheses are violated (Table 5, Figure 5). If $A \cap B$ is empty the union is disconnected; if it has one contractible component the union is acyclic; if it has three, $\tilde{H}_1 = \mathbb{Z}^2$; if A is a ball with a sealed cavity, H_2 appears. These are exactly the events that a donut-to-mug deformation must avoid, and they are what the program's measurements are designed to detect.

4.5 The family of shapes used by the program

Defining functions. For $A \subset \mathbb{R}^3$ closed and $\varepsilon > 0$ let $\operatorname{tube}(A, \varepsilon) = \{p : \operatorname{dist}(p, A) \le \varepsilon\}$; the function $\operatorname{dist}(\cdot, A) - \varepsilon$ is negative inside it, zero on its boundary and positive outside. The program describes each solid by such a function (a signed-distance-like function), and each stage t by one function F_t whose interior $S_t = \{F_t < 0\}$ is the solid.

The ring and its two arcs. For a centre abscissa m and radius R let $\Gamma(m, R) = \{(m + R\cos\varphi, 0, R\sin\varphi)\}$. For $p = (x, y, z)$ put $X = x - m$, $Z = |z|$, $\varrho = \sqrt{X^2 + Z^2}$ and let $\omega(p) \in [0, \pi]$ be the polar angle of (X, Z). With

$$\mathrm{d}_\Gamma(p) = \sqrt{(\varrho - R)^2 + y^2}, \qquad \mathrm{d}_\beta(p) = \sqrt{(X - R\cos\beta)^2 + (Z - R\sin\beta)^2 + y^2}$$

(the distances to the circle and to the endpoint $e(\beta) = (R\cos\beta, R\sin\beta)$ and its mirror image), and for $0 < \beta \le \pi$ the arcs $A(\beta) = \{\Gamma(\varphi) : |\varphi| \le \beta\}$ and $A^{\mathrm{c}}(\beta) = \{\Gamma(\varphi) : \beta \le |\varphi| \le \pi\}$, one has

$$\operatorname{dist}(p, A(\beta)) = \begin{cases} \mathrm{d}_\Gamma(p) & \omega(p) \le \beta \\ \mathrm{d}_\beta(p) & \omega(p) > \beta \end{cases}, \qquad \operatorname{dist}(p, A^{\mathrm{c}}(\beta)) = \begin{cases} \mathrm{d}_\Gamma(p) & \omega(p) \ge \beta \\ \mathrm{d}_\beta(p) & \omega(p) < \beta \end{cases} \tag{3}$$

because the nearest point of an arc to p is either the radial projection of p onto the circle, if that lies on the arc, or an endpoint. (The program tests $\omega \le \beta$ without trigonometry, by the sign of $\cos\beta\, Z - \sin\beta\, X$.) With the handle angle α, the overlap angle $\eta \ge 0$ and the tube radius r define

$$d_H = \operatorname{dist}(\cdot, A(\alpha)) - r, \qquad d_R = \operatorname{dist}(\cdot, A^{\mathrm{c}}(\alpha - \eta)) - r,$$

the defining functions of the handle tube T_h and of the tube T_r around the rest of the ring, which reaches past the handle's endpoints by the overlap angle η.

Cup and pit. With $\varrho_c(p) = \sqrt{(x - x_c)^2 + y^2}$ the distance from the cup's axis, let d_C be the defining function of the rounded cylinder $\operatorname{tube}(C_0, \varepsilon_c)$, $C_0 = \{\varrho_c \le R_c - \varepsilon_c,\ z_0 + \varepsilon_c \le z \le z_1 - \varepsilon_c\}$, and d_P that of the rounded pit $\operatorname{tube}(P_0, \varepsilon_p)$, $P_0 = \{\varrho_c \le R_c - w - \varepsilon_p,\ z \ge z_p + \varepsilon_p\}$, a cylinder open to the top whose floor sits at height z_p. (Here w is the wall thickness.)

Smooth union and intersection. For $k > 0$ put $h = \max(k - |a - b|, 0)/k$ and

$$\operatorname{smin}(a, b; k) = \min(a, b) - \tfrac{k}{4}h^2, \qquad \operatorname{smax}(a, b; k) = -\operatorname{smin}(-a, -b; k),$$

and $\operatorname{smin}(a, b; 0) = \min(a, b)$. Thus smin $\le$ min with equality when $|a - b| \ge k$: the sublevel set of $\operatorname{smin}(a, b; k)$ contains the union of the sublevel sets and differs from it only near the junction, where it adds a fillet. Dually smax $\ge$ max rounds a crease.

Table 2. The three phases of the family and their meaning for the decomposition of Figure 2.

schedule	window	what moves, and its role in the handle decomposition
σ_1	$0 \le t \le 0.6$	the ring contracts ($R : 1 \to 0.55$) and thins ($r : 0.36 \to 0.15$) and its handle arc opens from the full circle to $\lvert\varphi\rvert \le 112°$: the 1-handle becomes shorter and thinner
σ_2	$0.08 \le t \le 0.7$	the rest of the ring swells into the cup (blend weight s): the 0-handle grows
σ_3	$0.5 \le t \le 0.92$	the floor of the pit descends from 0.5 above the rim to 0.18 above the base: a dent is pushed into the 0-handle (Lemma 4.1)

The family. The stage-t function is

$$F_t = \operatorname{smax}\Big(\operatorname{smin}\big(d_H,\ (1-s)\,d_R + s\,d_C;\ k\big),\ -d_P;\ \kappa\Big), \qquad S_t = \{F_t < 0\}, \qquad \kappa = 0.05, \quad (4)$$

which in the program is the three lines

```
double d = smin(dHandle, mix(dRest, dCup, S.s), S.k);  // handle U body
...
return smax(d, -dPit, 0.05);                            // ... minus pit
```

The parameters depend on t through three smoothstep schedules $\operatorname{sm}(a, b; t)$ (equal to 0 for $t \le a$, to 1 for $t \ge b$ and to $3u^2 - 2u^3$ with $u = (t-a)/(b-a)$ in between):

$$\begin{aligned}
&\sigma_1 = \operatorname{sm}(0, 0.6; t), && \sigma_2 = \operatorname{sm}(0.08, 0.7; t), && \sigma_3 = \operatorname{sm}(0.5, 0.92; t),\\
&m = \sigma_1 m_1, && R = (1-\sigma_1)R_0 + \sigma_1 R_1, && r = (1-\sigma_1)r_0 + \sigma_1 r_1,\\
&\alpha = (1-\sigma_1)\pi + \sigma_1\alpha_1, && \eta = 0.7(1-\sigma_1), && s = \sigma_2,\\
&k = 0.22\ \operatorname{sm}(0, 0.10; \sigma_2), && z_p = (1-\sigma_3)(z_1 + 0.5) + \sigma_3 z_f. &&
\end{aligned} \quad (5)$$

The numerical values of all constants are collected in Appendix A. The schedules are monotone and constant outside their windows; they decide *when* things happen but not what the stage S_t is, which depends only on the parameter vector $(m, R, r, \alpha, \eta, s, k, z_p)$. Table 2 describes the three phases, Figure 3 shows the animation and Figure 4 the same family seen in the cross-section $y = 0$.

Observation 4.5 (the end shapes; numerical)**.** (a) At $t = 0$ the parameters are $\alpha = \pi$, $s = 0$, $k = 0$ and the pit lies above everything, so S_0 should be exactly the donut T. On 2×10^6 random points the sign of F_0 agreed with the sign of $\operatorname{dist}(p, \Gamma) - r$ in every case. (b) At $t = 1$ the seven probe points of Table 3 have the signs of the intended mug: material in the wall, floor and handle, empty inside the pit, above the rim and inside the window. The window is 0.536 wide.

The handle decomposition along the family. For $t > 0$ the solid S_t is the union of the (smoothed) handle tube $T_h(t)$ and a body $B_t = \{(1-s)d_R + s\,d_C < 0\}$, followed by the pit. The body is a blend between the thin tube T_r around the rest of the ring ($s = 0$) and the cup ($s = 1$); it grows from a bead on the left of the ring to the cup, while T_h shortens from the whole ring to the C-shaped handle. By Lemma 4.4, if every B_t is a ball and $T_h(t) \cap B_t$ has exactly two contractible components, then every S_t has Betti numbers $(1, 1, 0)$; the pit, by Lemma 4.1,

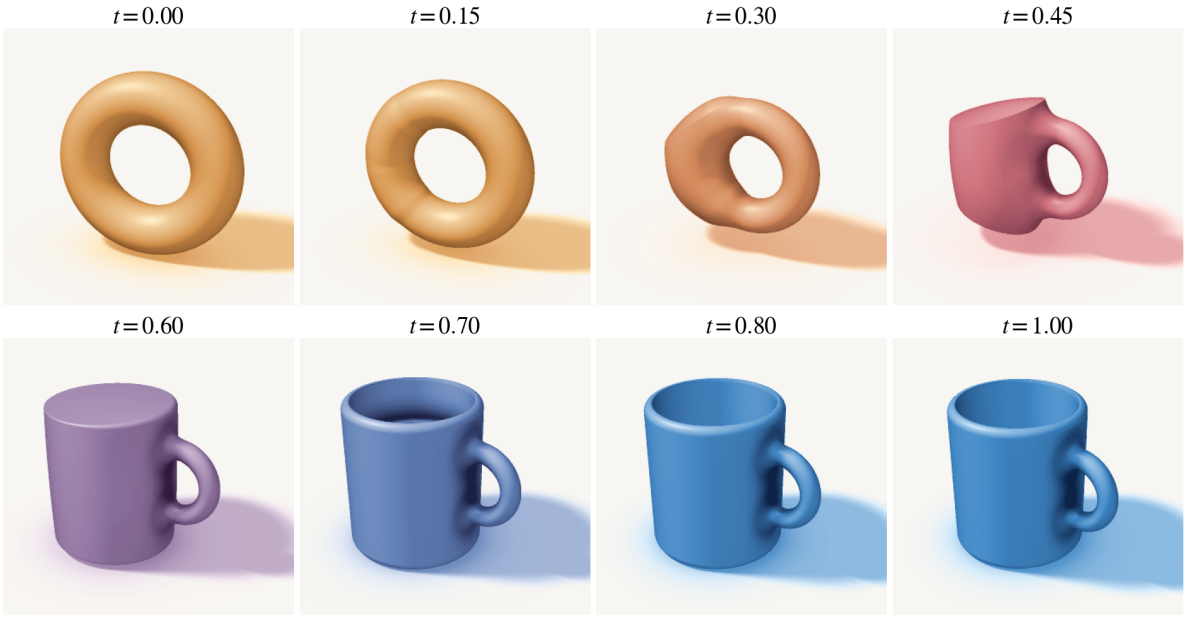


Figure 3. Eight frames of the animation produced by the program (rendered by ray marching; colour, lighting and camera play no role in the topology).

Table 3. Signs of F_1 at probe points of the final mug ($x_c = -1.30$, $R_c = 1.10$, $z_0 = -1.36$, $z_1 = 1.14$, $m_1 = -0.064$, $R_1 = 0.55$).

probe	point	F_1	meaning
centre of the cavity	$(x_c, 0, 0)$	$+0.960$	empty (inside the pit)
wall, handle side	$(x_c + R_c - 0.07, 0, 0)$	-0.057	material
middle of the floor	$(x_c, 0, z_0 + 0.09)$	-0.078	material
handle, centre of the tube	$(m_1 + R_1, 0, 0)$	-0.150	material
inside the window	$(m_1 + 0.2, 0, 0)$	$+0.192$	empty (the finger hole)
above the rim	$(x_c, 0, z_1 + 0.2)$	$+0.960$	empty
wall, side away from the handle	$(x_c - R_c + 0.07, 0, 0)$	-0.057	material

only deforms the ball B_t. We do not prove these two properties for all t; they are what the measurements below test.

Observation 4.6 (the ring does not tear; numerical)**.** For $t \in [0, 0.5]$, before the pit appears, all of 4.02×10^6 random points of the *full* ring tube $\mathrm{tube}(\Gamma_t, r(t))$ lie in S_t (the maximum of F_t over them is -1.5×10^{-7}). This is the quantitative form of "nothing is torn" in the first phase. It rests on two facts: $S_t \supseteq T_h(t)$ always (since smin $\leq$ min), and the part of T_r beyond the handle's angular range lies inside the cup: among 2.4×10^6 random points (401 values of t) there are 15 500 in T_r, of which 173 lie outside the cup, all of them with polar angle at most α—that is, in the stretch of T_r that overlaps the handle's own angular range and is already covered by T_h—and none beyond it. The largest excursion outside the cup is 0.078 length units.

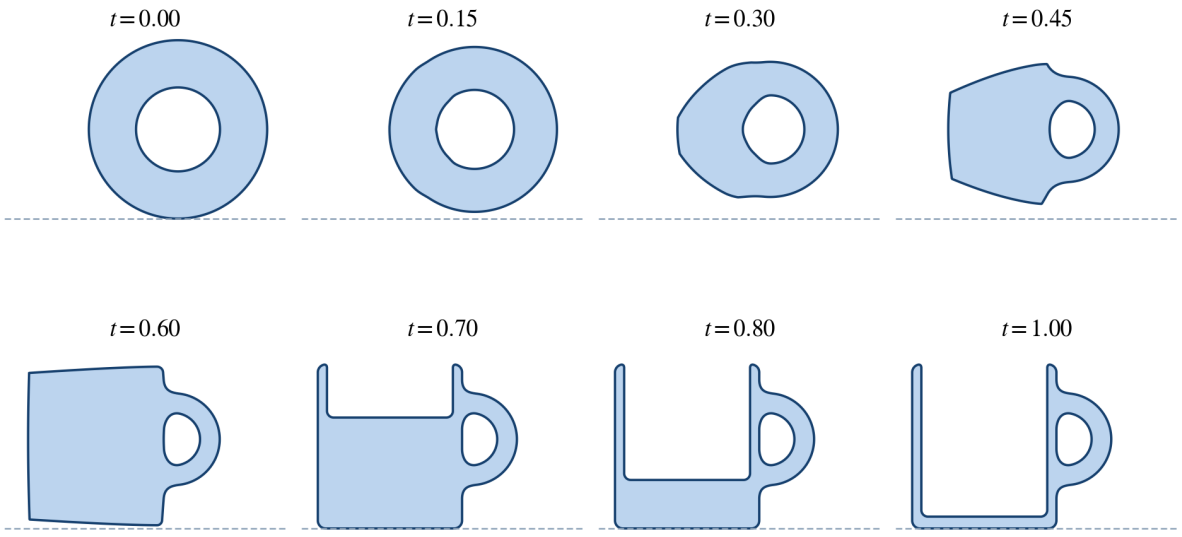


Figure 4. The cross-section $y = 0$ of S_t, computed from the program's own function F_t (shaded: $F_t < 0$; dashed: the floor). This plane contains the core loop, so the hole of the donut is visible as the hole of the section; during the deformation it becomes the window of the handle. At every one of these eight stages the section has one component and one hole.

4.6 When could the topology change?

Lemma 4.4 and Table 5 list the combinatorial events. Analytically, the cause of such an event is a critical point on the zero level set. The following is the standard level-set isotopy lemma (compare Milnor [8, Thm. 3.1] and Hirsch [3, Ch. 8]).

Proposition 4.7 (Level-set isotopy; classical)**.** *Let $F\colon \mathbb{R}^3 \times [0,1] \to \mathbb{R}$ be of class C^2 with $\{F_t \le 0\}$ contained in a fixed compact set for all t, and suppose $\nabla_p F(p,t) \neq 0$ whenever $F(p,t) = 0$. Then there is an ambient isotopy of diffeomorphisms Φ_t with $\Phi_t(\{F_0 < 0\}) = \{F_t < 0\}$. In particular all the sets $\{F_t < 0\}$ are diffeomorphic.*

Proof. Near the zero set put $X(p,t) = -\big(\partial_t F / |\nabla_p F|^2\big)\nabla_p F$, a C^1 field, cut it off smoothly away from the zero set and let Φ_t be its flow, which exists for all $t \in [0,1]$ since the field has compact support. For p with $F(p,0) = 0$ we get $\frac{d}{dt}F(\Phi_t(p),t) = \partial_t F + \nabla_p F \cdot X = 0$, so Φ_t maps $\{F_0 = 0\}$ into $\{F_t = 0\}$, and by running the flow backwards, onto. For p off the zero set $F(\Phi_t(p),t)$ never vanishes, so it keeps the sign of $F(p,0)$. □

Remark 4.8. The program's F_t is not C^2: it is built from minima, maxima, absolute values and square roots, and the smooth union and intersection are only C^1. Proposition 4.7 therefore does not apply verbatim, and we do not claim it. It explains the *mechanism*, though: the topology of $\{F_t < 0\}$ can change at a time t^* only if the zero level set has a critical point there—two sheets touching, such as the handle's inner side touching the wall (the window closes), the floor of the pit reaching the base, or the handle's end leaving the wall. These are exactly the events of Table 5, and the Betti-number computation is a detector for them.

4.7 What the program computes: Betti numbers of a voxel model

For each stage the program samples the sign of F_t at the centres of the voxels of a grid of spacing $h = 0.04$ covering the box $[-2.7, 1.6] \times [-1.5, 1.5] \times [-1.6, 1.6]$ ($107 \times 75 \times 80$ voxels), calls Ω_t the set of voxels whose centre has $F_t < 0$, and works with the *cubical complex* $K_t = K(\Omega_t)$: all vertices, edges, squares and cubes that are faces of at least one voxel of Ω_t. Its underlying space $|K_t|$ is the union of the closed voxels.

Theorem 4.9 (what the algorithm computes)**.** *Let Ω be a finite set of unit voxels in a box whose outermost layer of voxels is empty, and let $K = K(\Omega)$. Then*

(a) $\chi(K) = V - E + F - C$, where V, E, F, C are the numbers of vertices, edges, squares and cubes of K; the program's loop over the lattice corners counts every cell of K exactly once;

(b) $\beta_0(K)$ is the number of connected components of Ω under 26-adjacency (voxels sharing a face, an edge or a vertex);

(c) $\beta_2(K)$ is the number of connected components of the set of empty voxels under 6-adjacency (voxels sharing a face), minus one;

(d) $\beta_3(K) = 0$, hence $\beta_1(K) = \beta_0(K) + \beta_2(K) - \chi(K)$.

Proof. (a) A cell belongs to K iff it is a face of some voxel of Ω: a vertex iff one of the 8 voxels around it is in Ω, an edge iff one of the 4 around it, a square iff one of the 2, a cube iff it is in Ω. The program loops over all lattice corners and, at each, tests these conditions for the vertex at the corner, the three edges and three squares that start there in the positive directions, and the cube that starts there; every cell has exactly one such lowest corner. The Euler–Poincaré formula gives $\chi(K) = \sum_k (-1)^k \#\{k\text{-cells}\} = \sum_k (-1)^k \beta_k$ (Hatcher [1, §2.2]).

(b) Two voxels lie in the same component of $|K|$ iff they are joined by a chain of voxels of Ω in which consecutive voxels intersect, i.e. share a face, an edge or a vertex.

(c) K is a compact, locally contractible subspace of $S^3 \supset \mathbb{R}^3$. By Alexander duality (Hatcher [1, §3.3]), $\tilde{H}_0(S^3 \setminus K) \cong \tilde{H}^2(K)$, so $\beta_2(K)$ is the number of components of $S^3 \setminus K$ minus one; adding the point at infinity does not merge components of $\mathbb{R}^3 \setminus |K|$, and the empty outer layer of the box makes the exterior a single component. The set $\mathbb{R}^3 \setminus |K|$ consists of the open empty voxels together with the faces, edges and vertices that belong to no voxel of Ω. Two empty voxels sharing a face lie in the same component, since that face belongs to no voxel of Ω. Two empty voxels sharing only an edge or a vertex can be joined through it only if all the voxels around that edge or vertex are empty, in which case they are linked by face-adjacent chains anyway. So the components of the complement are the 6-connected components of the empty voxels.

(d) $H_3(K) = 0$: a non-zero 3-cycle would have a cube of maximal height with non-zero coefficient, whose top face would then occur in the boundary, which is a contradiction. The last formula is (a) with $\chi = \beta_0 - \beta_1 + \beta_2$. □

Remark 4.10. The pairing (26 for the shape, 6 for its surroundings) is forced by using *closed* cubes. Using the same adjacency for both leads to the familiar paradoxes of digital topology: in the plane, a 2×2 checkerboard of pixels has, under 8-adjacency for both colours, a connected foreground and a connected background crossing each other (a Jordan-curve paradox), and under 4-adjacency for both, neither colour connected. See Kong and Rosenfeld [6] and Kaczynski–Mischaikow–Mrozek [5].

Table 4. The voxel complex K_t ($h = 0.04$): numbers of vertices V, edges E, squares F and cubes C, and the resulting $\chi = V - E + F - C$ and Betti numbers $(\beta_0, \beta_1, \beta_2)$. The program checks 41 stages; every fourth is shown.

t	V	E	F	C	χ	$(\beta_0, \beta_1, \beta_2)$
0.0	45 852	130 942	124 526	39 436	0	$(1, 1, 0)$
0.1	41 566	118 406	112 308	35 468	0	$(1, 1, 0)$
0.2	36 390	103 518	98 049	30 921	0	$(1, 1, 0)$
0.3	32 532	92 525	87 629	27 636	0	$(1, 1, 0)$
0.4	43 102	123 765	118 399	37 736	0	$(1, 1, 0)$
0.5	77 576	225 508	218 479	70 547	0	$(1, 1, 0)$
0.6	131 118	383 968	374 794	121 944	0	$(1, 1, 0)$
0.7	125 390	363 904	351 836	113 322	0	$(1, 1, 0)$
0.8	84 376	238 634	224 314	70 056	0	$(1, 1, 0)$
0.9	61 886	169 949	154 401	46 338	0	$(1, 1, 0)$
1.0	61 794	169 672	154 123	46 245	0	$(1, 1, 0)$

4.8 Results

Observation 4.11 (Betti numbers along the family; numerical)**.** The program computes $(\chi; \beta_0, \beta_1, \beta_2)$ for all 41 stages $t = 0, 1/40, \ldots, 1$ and flags any stage that is not $(1, 1, 0)$. None is flagged. Table 4 shows every fourth stage together with the cell counts; for example at $t = 0$, $45\,852 - 130\,942 + 124\,526 - 39\,436 = 0$, and at $t = 1$, $61\,794 - 169\,672 + 154\,123 - 46\,245 = 0$.

Observation 4.12 (robustness; numerical)**.** The result does not depend on the particular voxel grid: the 41 stages gave $(1, 1, 0)$ for each of the voxel sizes $h = 0.03, 0.04, 0.05$, each with the grid unshifted and shifted by 0.0137; 161 equally spaced stages ($h = 0.04$, shifted grid) and 401 equally spaced stages ($h = 0.05$) gave no deviation either.

Observation 4.13 (controls; numerical)**.** To check that the measurement can actually fail, the program's own routine was applied to deliberately broken mugs (and one broken donut). In every case the computed values equal those predicted by the Mayer–Vietoris computation of Section 4.4; see Table 5 and Figure 5.

4.9 How far the evidence goes

1. **Voxel model versus smooth shape.** K_t is an approximation of S_t. Sampling theorems for digitizations of "r-regular" shapes (shapes whose boundary has bounded curvature on both sides, so that balls of radius r roll freely inside and outside) say that a sufficiently fine grid preserves the homotopy type under the (26,6) pairing; see Pavlidis [10], Latecki–Conrad–Gross [7] and Stelldinger–Köthe [11]. The thinnest features here are the cup's wall (thickness 0.14, 3.5 voxels), its floor (0.18) and the handle (diameter 0.30), so the hypothesis is not verified with comfortable margin, which is why Observation 4.12 repeats the test at other spacings and offsets.
2. **Finitely many stages.** The Betti numbers are piecewise constant in t. A topology change that began and was undone between two consecutive stages (a birth–death pair of critical points) would be invisible. Testing 401 stages makes this unlikely, not impossible.

Table 5. Controls: prediction by Lemma 4.4 and its variants versus the program's own computation of $(\chi; \beta_0, \beta_1, \beta_2)$. Here $S = A \cup B$ with A the cup and B the handle.

situation	structure of $A \cap B$	predicted	computed	
the mug (normal case)	two contractible pieces	$0; (1,1,0)$	$0; (1,1,0)$	✓
handle removed ($r = 0$)	$S = A$ is a ball	$1; (1,0,0)$	$1; (1,0,0)$	✓
handle so fat that the window closes	one contractible piece	$1; (1,0,0)$	$1; (1,0,0)$	✓
handle moved away (detached)	empty	$2; (2,0,0)$	$2; (2,0,0)$	✓
thin strut across the window	three contractible pieces	$-1; (1,2,0)$	$-1; (1,2,0)$	✓
cup sealed by a lid	A has a sealed cavity ($\beta_2(A) = 1$)	$1; (1,1,1)$	$1; (1,1,1)$	✓
donut whose tube is so fat that the hole closes	(no handle decomposition)	$1; (1,0,0)$	$1; (1,0,0)$	✓

3. **Invariants do not classify.** A knotted solid torus also has Betti numbers $(1,1,0)$, and homology cannot distinguish the solid torus from the solid Klein bottle. The check cannot see knotting; that the family is an honest deformation of the donut into the mug rests on its construction and on the pictures (Remark 2.2), not on the Betti numbers.

4. **No proof for all t.** We have proved the two end shapes are homeomorphic (Proposition 4.3) and proved what the algorithm computes (Theorem 4.9). We have *not* proved that every S_t is a solid torus; the evidence is Observations 4.6 and 4.11 to 4.13. A proof would need either interval arithmetic or an analytic check that no critical point occurs on the zero level set of F_t for any t (Remark 4.8).

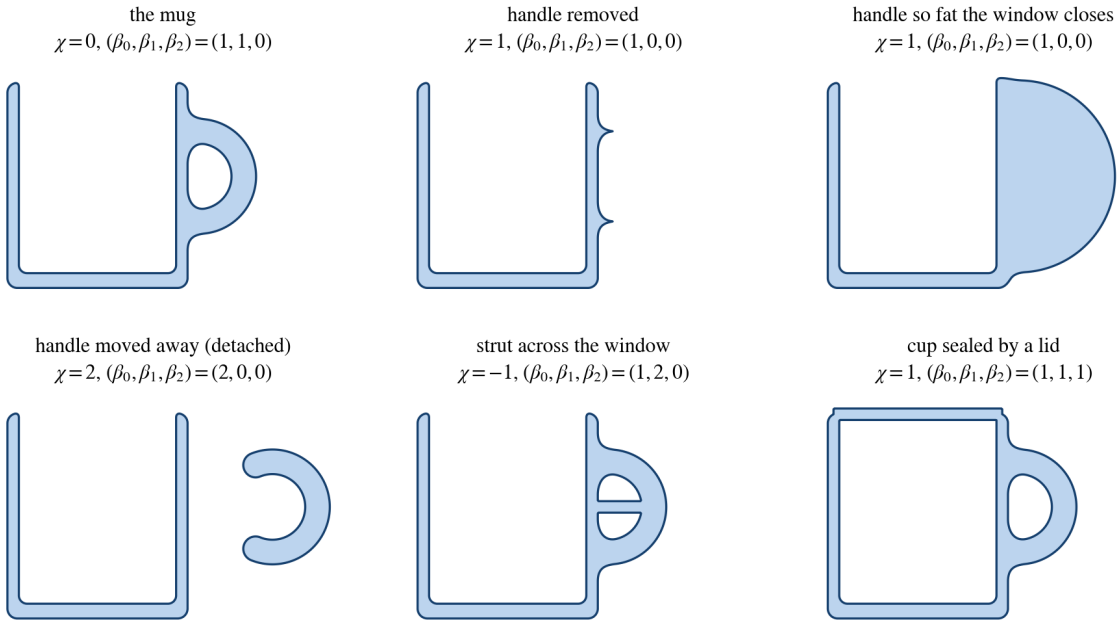


Figure 5. The control experiments of Table 5 in the cross-section $y = 0$, with the computed Euler characteristic and Betti numbers of the *whole* solid. A planar section is only an illustration (for instance the sealed cavity of the lid appears as a second hole of the section, whereas in the solid it contributes to β_2, not β_1).

5 A dictionary: from code to mathematics

Table 6. What the identifiers in the two programs mean in the language of this note.

code	mathematical object
Program I	
`circleToSquare(p)`	the homeomorphism F of $\mathbb{R}^2$, (1); on S^1 it is $p \mapsto p/\|p\|_\infty$
`squareToCircle(q)`	its inverse G
`morph(p,t)`	the ambient isotopy H_t, (2)
`windsOnce(poly)`	discrete test that a closed polygon winds once around the origin with strictly increasing angle, hence is a simple closed curve (Corollary 3.3)
printed maximum errors	residuals of $G \circ F - \mathrm{id}$, $F \circ G - \mathrm{id}$ and of $F(S^1) \subset \Sigma$
Program II: the family	
`Shape`, `shapeAt(t)`	the parameter vector $(m, R, r, \alpha, \eta, s, k, z_p)$ as a path in t, (5)
`smooth(a,b,x)`	the smoothstep schedule $\mathrm{sm}(a, b; x)$
`dCircle`, `dEndH`, `dEndR`	d_Γ, d_α, $\mathrm{d}_{\alpha-\eta}$ of (3)
`dHandle`, `dRest`	d_H and d_R, the handle tube and the tube around the rest of the ring
`dCup`, `dPit`	d_C and d_P, the rounded cup and the pit
`smin`, `smax`	smooth union and intersection
`sdf(p,S)`	the defining function F_t, (4); the solid is $S_t = \{F_t < 0\}$

code	mathematical object
Program II: the topology check	
`measureTopology(S)`	$(\chi, \beta_0, \beta_1, \beta_2)$ of the cubical complex K_t (Theorem 4.9)
loop over lattice corners	the cell counts V, E, F, C
`components(want,diagonals)`	connected components: 26-adjacency for the shape, 6-adjacency for the empty space
No topological content	
`trace`, `renderFrame`, `cameraAt`, `makePalette`	ray marching, shading, camera and colours of the animation; the marching step uses only that F_t is 1-Lipschitz (Hart [4])
`lzwEncode`, `writeGif`	the GIF file format

6 Conclusion

The circle-to-square program is a theorem in disguise: the radial change of norm is a homeomorphism of the plane, and the straight line from the identity to it is an ambient isotopy, so the picture is exact and the corners are an artifact of the embedding, not of the topology. The donut-to-mug program is of a different kind. Its two ends are the same space for a classical reason, a 0-handle with one 1-handle, and its deformation moves from one such decomposition to another by growing the ball, shortening the handle and denting the ball. What the program adds to the theory is a measurement: at every stage it counts the pieces, tunnels and cavities of a voxel model, and the answer, one piece, one tunnel and no cavity, never changes, while broken variants change it exactly as the Mayer–Vietoris sequence predicts. Table 7 separates what has been proved from what has only been observed.

Table 7. Ledger of claims.

claim	status	where
F is a homeomorphism of $\mathbb{R}^2$ carrying circles onto squares	proved	Proposition 3.1
H_t is an ambient isotopy from the circle to the square	proved	Proposition 3.2
a dent with positive floor thickness does not change the ball	proved	Lemma 4.1
the solid donut and the mug are homeomorphic	classical, argued	Proposition 4.3
two balls glued along two discs have the homology of a solid torus	proved	Lemma 4.4
the voxel algorithm computes the exact Betti numbers of K_t	proved	Theorem 4.9
S_0 is exactly the donut; S_1 has the intended geometry	numerical	Observation 4.5
the ring is never torn before the pit appears	numerical	Observation 4.6
$(\beta_0, \beta_1, \beta_2) = (1, 1, 0)$ at 41 stages, and at 401 finer ones	numerical	Observations 4.11 and 4.12
the broken controls give the Betti numbers predicted by Lemma 4.4	numerical	Observation 4.13
every S_t, $t \in [0, 1]$, is a solid torus	**not proved**	Section 4.9
the voxel models have the topology of the smooth solids	**not proved**; supported	Section 4.9

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A Parameter values

Table 8. Constants of Program II (names as in the source file).

symbol	value	meaning (source name)
$R_0,\ r_0$	$1.00,\ 0.36$	donut: ring radius and tube radius (`DONUT_R`, `DONUT_TUBE`)
$x_c,\ R_c$	$-1.30,\ 1.10$	cup: axis abscissa and outer radius (`CUP_X`, `CUP_RAD`)
$z_0,\ z_1$	$-1.36,\ 1.14$	base ($= -(R_0 + r_0)$, where the donut's bottom was) and rim (`CUP_Z0`, `CUP_Z1`)
$\varepsilon_c, w, z_f - z_0, \varepsilon_p$	$0.12, 0.14, 0.18, 0.10$	edge rounding, wall thickness, floor thickness, pit rounding (`CUP_ROUND`, `WALL`, `FLOOR_T`, `PIT_ROUND`)
$R_1,\ r_1,\ \alpha_1$	$0.55,\ 0.15,\ 112^\circ$	handle: ring radius, tube radius, half-angle of the arc
m_1	-0.0640	handle ring centre: $(x_c + R_c - 0.07) - R_1 \cos\alpha_1$, so that the handle's ends lie 0.07 inside the wall
$\eta_0,\ k_{\max},\ \kappa$	$0.7,\ 0.22,\ 0.05$	overlap angle (rad), fillet radius, rim-rounding radius
schedules	see (5)	σ_1 on $[0, 0.60]$, σ_2 on $[0.08, 0.70]$, σ_3 on $[0.50, 0.92]$
voxels	$h = 0.04$	box $[-2.7, 1.6] \times [-1.5, 1.5] \times [-1.6, 1.6]$; 41 stages

B How the numbers were produced

All figures and tables were generated from the delivered programs. Small test harnesses include the source file of Program II up to `main()` and call its functions (`shapeAt`, `sdf`, `measureTopology`) directly: they dump the cross-sections of F_t used in Figures 4 and 5, the cell counts of Table 4, the probe values of Table 3 and the random-sample statistics of Observations 4.5 and 4.6. The control shapes of Table 5 are obtained by changing a parameter of the final stage ($r = 0$; $r = 0.62$; $m \mapsto m + 0.9$) or by adding to F_1 a lid (a disc of radius 1.03 and thickness 0.14 on the rim) or a strut (a bar of radius 0.07 along the x-axis across the window). Program I was rerun from its delivered file for the numbers of its section. Every row of Table 4 satisfies $V - E + F - C = \chi$ (re-verified when the table was generated), and every control outcome equals its prediction.